

Decoupling Workload Admission from Policy Materialization in Kubernetes

*AI-generated working draft for author discussion
Non-submission exercise; Traditional Research framing*

Editorial notice. This is an AI-generated writing exercise, not a submission or text attributed to Antonio Ojea. It develops the proposed research argument using the available evidence. Implementation inspection, archived measurements, operator-reported outcomes, and analytical predictions are distinguished throughout. The evaluation is incomplete.

Abstract

Network policy enforcement must keep pace with the creation and replacement of the workloads it protects. Indexed kernel lookups and incremental datapath updates improve packet processing, but they do not by themselves remove the cost of translating changing workload metadata into distributed enforcement state. We examine a different placement of that work: evaluate Kubernetes policy semantics in userspace when communication is attempted, and reuse accepted decisions through the kernel’s connection tracker. The kube-network-policies implementation combines NFQUEUE-based admission with nftables packet steering; runtime integration provides local Pod metadata without waiting for its asynchronous publication through the Kubernetes API. We characterize the design using source inspection and archived experiments targeting 35,000 Pods on a reported 720-worker testbed. Completed Cilium identity sweeps retain low post-scheduling latency and flat worker CPU (0.51–0.63 mean cores) across 7 to 35,000 intended identities, while the 35,000-ReplicaSet tiers halve achieved Pod throughput (170 versus 280–330 Pods/s) and a dense-mesh tier was excluded because its estimated per-endpoint permission state exceeded the stated map capacity. An earlier Cilium release stranded Pods at 700 identities, and a separate direct-burst experiment stranded 5,694 Pods. These observations motivate distinguishing latency, throughput, capacity, and admission failure, rather than treating identity cardinality as a single scaling variable. We identify the conditions under which deferred evaluation reduces eager policy state and evaluate a kube-network-policies bidirectional mesh sweep that matches Cilium’s at three ReplicaSet-paced tiers (7, 700, and 3,500 identities) and adds an unmatched 35,000-identity raw-Pod tier.

Because every sandbox blocks in an init container until a permitted TCP connection to a gateway succeeds, the reported post-scheduling latency bounds time to first permitted communication, including cross-node metadata convergence. At 700 and 3,500 identities kube-network-policies reaches 5.33 s and 2.79 s P99, within 2.4 s of Cilium at 2.5–3× lower achieved Pod throughput; at 7 identities with 5,000 Pods per ReplicaSet its P99 is 151 s against Cilium’s 6.8 s under the same generator; and the 35,000 raw-Pod tier, which Cilium’s producer excluded analytically, completes with all Pods Running but a 421 s P99 whose cause the artifacts do not isolate. We analyze the connection-processing and fail-open trade-offs of deferred evaluation and enumerate the measurements still needed to complete the comparison.

1 Introduction

The unit of isolation in a container cluster is also a unit of change. A service scales out, a job completes, or a sandbox is replaced; its network permissions must follow that lifecycle. For long-lived services, much of this work can be amortized over hours of communication. Short-lived workloads offer less time to recover their setup cost. Agent execution environments provide one motivation for this regime, alongside batch jobs and serverless functions, although the experiments considered here are synthetic sandbox workloads rather than traces of deployed agents.

Kubernetes makes the problem more subtle than installing a firewall rule for a new address. NetworkPolicies describe permitted relationships through Pod labels, namespace labels, ports, and address ranges. Addresses and placement change independently of this intent. An implementation must join policy with current workload metadata, establish isolation at the appropriate point in the workload lifecycle, and handle updates while packets continue to arrive. A fast packet lookup is necessary in many settings, but it is not a complete answer to these coordination costs.

Successive enforcement mechanisms have addressed different parts of this problem. Linear chains give way to indexed

sets and maps. Programmable datapaths separate packet-processing code from mutable state. Distributed network virtualization compiles logical intent into local actions and caches expensive decisions. These mechanisms are complementary, not a sequence of obsolete technologies. In particular, nftables supports incremental updates, and userspace miss handling followed by kernel caching is central to Open vSwitch [9, 26]. The next question is therefore not which kernel mechanism is newest. It is which work must occur before a workload can communicate, and how much of that work must be repeated elsewhere in the cluster.

This paper examines *deferred semantic evaluation* for Kubernetes network policy. In kube-network-policies, hereafter KNP, a node agent receives packets selected by nftables through NFQUEUE, resolves the relevant metadata, and evaluates the policy in userspace. An accepted verdict records an acceptance label in conntrack. Eligible subsequent traffic follows the kernel path without repeating the semantic evaluation. The engine does not require a separate policy-specific distributed security-identity allocator or a per-endpoint expansion of every permitted peer identity. It still requires metadata distribution, packet steering, and per-connection state.

The design trades eager permission materialization for work at attempted communication. If a policy permits a large peer set but a workload contacts only a few peers, maintaining every possible relationship may be unnecessary. Conversely, a workload creating many short connections can make the queue and userspace evaluator the bottleneck. Denied traffic is especially important: repeatedly rejected attempts do not benefit from the accepted connection cache. The research question is whether this tradeoff is favorable for high-churn clusters, not whether userspace processing is universally faster than in-kernel processing.

Local startup observation is a second part of the design. Kubernetes publishes a Pod’s address asynchronously. A policy agent relying exclusively on that publication can learn the address after a container has started using it, or after a lifecycle hook has begun waiting for network access. The longstanding issue documented in Kubernetes issue 85966 illustrates this dependency [11]. Node Resource Interface (NRI) integration obtains local sandbox metadata through the runtime, removing that particular round trip from local address discovery. It does not eliminate namespace or policy convergence, nor establish a cluster-wide startup barrier.

Our evidence distinguishes five observations. First, the completed Cilium identity sweeps do not show an increasing post-scheduling latency curve: the intended 35,000-identity open-gateway configuration has a 1.99-second P99, and mean worker CPU stays within 0.51–0.63 cores across all tiers. Second, the 35,000-ReplicaSet tiers reduce achieved Phase-1 throughput from roughly 280–330 to 170 Pods/s and raise cluster-scoped Pod LIST P99 latency from 14–19 s to 28.8 s; the controller fan-out, not the datapath, is the visible bottleneck there. Third, under the benchmark’s per-endpoint ex-

pansion model, a bidirectional mesh with 35,000 identities requires approximately 70,000 peer-policy entries per endpoint. The experiment producer excluded that tier on predicted capacity grounds; it was not executed on Cilium. Fourth, two Cilium admission failures are recorded: an archived Cilium v1.18.6 run stranded 19 of 35,000 Pods at 700 identities, attributed to a policy-cache race fixed in v1.20.0 [1], and a v1.20.1 raw-Pod burst aborted with 29,306 Pods Running and 5,694 stranded, attributed to endpoint throttling and cumulative identity allocation. Fifth, a KNP bidirectional mesh sweep matched to Cilium’s at the three ReplicaSet-paced tiers shows no increase in post-scheduling P99 from 700 to 3,500 identities (5.33 s to 2.79 s), but a 151 s P99 at the 5,000-Pods-per-ReplicaSet tier where Cilium records 6.8 s, and an unmatched 35,000-identity raw-Pod tier that completes with all Pods Running at a 421 s P99. The KNP runs also achieved $2.5\text{--}3\times$ lower Pod throughput than Cilium’s, so the sweep does not by itself separate identity cost from generator pacing. None of these observations should be substituted for another.

The paper develops three contributions at different levels of evidence. It describes a working enforcement decomposition based on existing Linux mechanisms, including a security analysis of its fail-open and unknown-peer behavior; presents a conditional model of eager permission state versus connection-driven work; and analyzes an archived large-cluster benchmark collection across both Cilium and KNP, including the newly executed KNP mesh sweep. Section 6.7 enumerates what the artifacts still do not contain: the KNP sweep has one trial per tier, its agent-level metrics returned no samples, its 35,000-identity tier has no Cilium counterpart under the same policy, and no denied-traffic or overload measurement exists. Making those limits explicit lets the evaluation test the design rather than assuming its conclusion.

2 Background and Problem

2.1 Policy Semantics and Workload Lifecycle

Kubernetes NetworkPolicy isolates ingress and egress independently. Within a direction, the permissions of applicable standard NetworkPolicies combine additively. A connection between two isolated Pods must satisfy the source’s egress policy and the destination’s ingress policy; reply traffic for an allowed connection is implicitly permitted [19]. This connection-oriented interpretation makes reuse of admission decisions natural, although it does not require any particular conntrack implementation.

Policy creation and workload creation have different synchronization requirements. A newly submitted policy is handled asynchronously, and the API does not expose a general enforcement-completion barrier. Once a plugin has handled a policy, newly created selected Pods must be isolated before their containers start. Allow permissions may converge later. Thus a system can preserve isolation while temporarily delay-

ing legitimate communication. Safety and availability must be measured separately; a fast transition to `Running` is not proof of either.

We distinguish four quantities that are often conflated: live endpoints, distinct policy-relevant label sets, allocated numeric security identities, and materialized datapath entries. Several Pods can share an identity. One identity can contribute permissions to many endpoint maps. Allocations not yet reclaimed can outlive their original workload. The number of permitted relationships also differs from the number of active connections. The resulting scaling dimensions are related but not interchangeable.

2.2 Placement and Timing of Enforcement Work

Chains, sets, and incremental updates. An ordered chain can perform work proportional to the number of rules visited. This does not imply that every iptables deployment performs a linear address lookup: ipset supplies indexed membership, while nftables integrates sets and verdict maps. Sutter’s 2017 measurements demonstrate the importance of this representation choice, with both nftables sets and iptables with ipset maintaining throughput across the tested ranges [12]. More recent Kubernetes work describes incremental nftables updates that scale with changed Service state rather than the full Service ruleset [26]. Service proxying is not NetworkPolicy, but it demonstrates why lookup and update mechanisms must be discussed independently.

Virtual networking and reactive caches. NVP and Open vSwitch address the translation from logical networks to physical forwarding [4, 9]. OVS caches flow-processing results and sends misses to userspace; its evolution illustrates both the value and the cost of reactive processing. OVN also supports stateful ACLs, including `allow-related` actions backed by connection tracking [23]. A comparison that describes these systems as inherently stateless would miss the closest precedents for our design.

Programmable datapaths and identities. eBPF maps permit userspace to update keyed state without replacing the packet-processing program [20]. Cilium uses label-derived security identities to represent workloads [15]. Identity sharing can avoid repeating work for equivalent workloads, but distinct labels and broad selectors can still create allocation, distribution, and materialization work. These are properties of an architecture and configuration, not intrinsic restrictions of eBPF. The size of a numeric identity space and the capacity of a policy map are different limits.

Problem statement. We ask whether policy enforcement can retain the established kernel forwarding path while avoid-

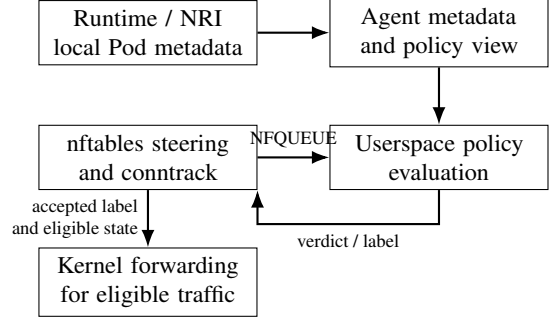


Figure 1: Enforcement decomposition, omitting protocol and host-traffic exceptions. Kubernetes watches also feed the agent’s policy and metadata view. NRI supplies local observations, not remote policy synchronization.

ing eager expansion of all permitted peers at each endpoint. A useful solution must explain where the displaced work goes, how local admission obtains sufficiently fresh metadata, and how existing decisions interact with changing policy. Packet throughput alone cannot answer these questions.

3 Design

3.1 Connection Admission and Cached Decisions

KNP separates policy interpretation from packet steering. Kubernetes objects supply policy intent and workload metadata to the node agent. nftables identifies traffic requiring a decision and sends it to NFQUEUE. The callback parses packet headers, invokes the userspace evaluator, and returns an accept or drop verdict. For an accepted decision it supplies a conntrack label, which the kernel rules can subsequently recognize. Figure 1 illustrates the separation.

The cached decision is connection state, not a precomputed table of every allowed peer. In the inspected forwarding path, the bypass rule requires both the acceptance label and an established or related conntrack state. Uncached managed traffic is queued. Consequently, “first-packet evaluation” is a useful intuition but not an exact accounting rule: several initial packets, retransmissions, or protocol-specific transitions can reach userspace before the bypass condition holds. Denied packets do not acquire the accepted-decision cache entry.

nftables remains part of the implementation. It maintains steering rules and managed-address sets, rather than executing Kubernetes label selectors as a shared kernel policy table. Avoiding peer-permission expansion does not imply eliminating every local rule update. Changes to managed endpoints or enforcement scope can still require steering reconciliation. The policy evaluator’s own metadata indexes and selector processing also consume CPU and memory.

The decision is local once the required metadata has arrived:

the packet path does not allocate a new distributed numeric identity for the attempted connection. This is a receive-driven policy metadata path, not a claim that the entire network control plane is read-only. The runtime, address management, routing, and Kubernetes controllers continue to update their own state. Remote endpoint metadata must still reach the enforcing node.

3.2 Runtime Metadata and Startup Isolation

Consider the dependency chain when a node learns its own Pod’s IP only through Kubernetes status. The runtime creates the sandbox and configures networking; the kubelet publishes status to the API server; a watch event reaches the policy agent. Container or lifecycle-hook activity can occur before the last step. If that activity waits for network permissions that depend on the status update, observation delay becomes application delay. Issue 85966 records precisely this problem for policy-only networking [11].

The NRI resolver populates a local IP-to-Pod metadata cache in its `RunPodSandbox` callback. It uses addresses supplied by the runtime API and can fall back to inspection of the sandbox network namespace. This places local observation at the runtime boundary instead of behind the kubelet–API-server–watch round trip. Subject to the runtime’s callback ordering and successful metadata acquisition, it makes the local address available before application-container execution requires it.

There are two limits to this improvement. First, obtaining metadata is not the same as installing effective isolation. The integration must ensure the steering path cannot bypass policy before protection applies. Second, namespace labels, policy objects, and remote peers can still be asynchronous. A claim of zero startup latency would therefore be incorrect. The narrower claim is removal of a specific asynchronous dependency in local address discovery; an NRI-on/off timing and connectivity experiment is needed to quantify its benefit.

3.3 Policy Changes and Failure Semantics

Caching makes policy changes part of the correctness argument. Standard `NetworkPolicy` leaves the effect of updates on already established connections implementation-defined [19]. KNP also provides a strict mode that scans conntrack entries, reevaluates supported flows, and clears the acceptance label where current policy no longer permits communication. Subsequent processing can then revisit the decision. This introduces work proportional to the flows inspected and the complexity of reevaluation; it is not an instantaneous revocation mechanism. The enforcement delay must include event delivery, scan scheduling, scan execution, and subsequent packet handling.

Failure policy must be equally explicit. `NFQUEUE` can drop packets on queue overflow or accept them when config-

ured fail-open [21]. The inspected command-line interface defaults to fail-open, and the dataplane sets the queue’s bypass flag whenever fail-open is enabled, with a maximum queue length of 1,024 packets. Two facts about the cached path make this a security-relevant default rather than an operational detail. Denied packets never acquire the acceptance label, so every retried forbidden connection attempt re-enters the queue; and a workload that can emit uncached packets faster than the agent returns verdicts fills the queue. Under fail-open, the overflow is then *accepted* by the kernel without evaluation. A tenant that is permitted to run code in the cluster can therefore attempt to convert denied traffic into a bypass by saturating its node’s agent, and its own denied attempts are the cheapest saturating traffic. The inspected implementation has no per-source rate limit or denied-flow negative cache that would blunt this amplification; the only countermeasure is `-fail-open=false`, which converts overload into loss of connectivity for every managed Pod on the node. Quantifying the overload point is therefore part of the safety argument, not only of the performance argument. The benchmark manifest in the repository sets `-nfqueue-id` but not `-fail-open`, so the archived KNP runs executed with the fail-open default.

Unknown peers have a different failure mode. When the evaluator cannot resolve a remote source or destination address to Pod metadata, the inspected selector matching treats a `nil` peer as matching only `ipBlock` rules; Pod and namespace selectors never match it. Under a default-deny policy the packet is dropped. This is fail-closed at the policy level, and it is the correct isolation behavior, but it means a permitted connection from a freshly created remote Pod is denied until the destination node learns the source’s metadata. NRI removes this delay only for *local* Pods. For remote peers the agent still depends on API-server watch delivery, or on the `IPTracker` distribution tier of Section 8.1, and the observed delay is an availability cost. The benchmark’s startup gate exercises exactly this path: a new sandbox’s first connection is admitted at the gateway node only after that node has learned the sandbox’s address, so the `schedule_to_run` latencies of Section 6 contain this convergence time rather than exclude it.

Further, the inspected implementation logs an initial `nftables` synchronization error without blocking startup, and its fail-closed graceful-shutdown path removes its `nftables` rules. Those transitions require a deployment-level isolation contract or implementation changes before claiming protection throughout the lifecycle.

We therefore state the desired safety property conditionally. For a policy already handled by an agent, a new selected workload must not send or receive forbidden traffic before enforcement takes effect. An allowed connection may be delayed by missing information; such delay must not be confused with an isolation violation. The current benchmark collection does not establish this property. A complete validation must exercise missing metadata, label changes, IP reuse, failures, and

init containers as well as steady-state allow/deny behavior.

The intended trust boundary includes the node kernel, runtime, policy agent, and the CNI’s endpoint/address binding. Untrusted workloads must not be able to impersonate a peer by spoofing its address. Privileged workloads, host networking, and enforcement around address translation need explicit scope. NFQUEUE and NRI do not independently establish those assumptions.

4 Where the Scaling Cost Moves

Let I denote selected peer identities, P_ℓ local endpoints, d the number of materialized directions, and F_ℓ locally tracked active connections. In an eager representation with a distinct entry per peer identity and direction at every endpoint, the peer-policy contribution is

$$E_{\text{endpoint}} \approx dI, \quad E_{\text{node}} \approx dIP_\ell. \quad (1)$$

This is a conditional representation model, not a lower bound on all policy implementations. Sharing, wildcarding, aggregation, or another representation can change it. Additional policy rules contribute further entries. Entry counts are also not byte counts: data-structure overhead, preallocation, and sharing must be measured separately.

Under the same assumptions, a new identity selected by every local endpoint can induce approximately dP_ℓ local entry updates. For a rate λ_I of such new identities, the corresponding update demand is approximately $dP_\ell\lambda_I$. Identity dissemination, selector recomputation, and endpoint regeneration are separate costs; none is measured by this expression. With a shared identity, many Pod replacements may leave this demand nearly unchanged. Fresh-identity churn is consequently a different workload from Pod churn.

Deferred evaluation removes this particular eager cross-product, but retains metadata, steering state, and conntrack. A useful accounting is

$$M_{\text{deferred}} = M_{\text{metadata}} + M_{\text{steering}} + M_{\text{conntrack}}(F_\ell). \quad (2)$$

It is not a proof that total memory is smaller than in an identity-based engine: both may already maintain conntrack, and the userspace metadata representation can dominate. The avoided term is materialized permission state; the comparison must account for all other state on both sides.

Let λ_q be the actual queued-packet rate and μ the aggregate sustainable verdict-processing rate. Avoiding sustained queue growth requires, as a first-order condition,

$$\rho = \lambda_q / \mu < 1. \quad (3)$$

Burstiness, shared CPU, and scheduling can produce tail latency and drops even below this average-load condition. Queue demand includes denied attempts and repeated initial packets, not just successfully established connections.

The only measurement of μ available to us is the repository’s own three-node ApacheBench microbenchmark [18]: 10,000 non-keepalive HTTP requests at concurrency 1,000 completed at 2,317 requests/s with a 5-ms median and a 3,080-ms P99 dominated by connect time, while the kernel logged `nf_conntrack: table full` at the default 262,144-entry limit. The accompanying Prometheus charts show per-node queued-packet peaks near 8,000 packets/s and callback durations with medians of 80–120 μs and P99s of 170–630 μs . Hardware, kernel, and agent version are not recorded, and conntrack exhaustion confounds the tail, so we treat these as an order-of-magnitude bound on a single node’s verdict capacity, not as a characterization of μ . The 720-node startup reports do not establish λ_q at all.

The favorable regime is thus not simply “many identities.” It is a large or rapidly changing set of possible permissions relative to the communication actually attempted, with sufficient admission-processing capacity. A dense *permission* graph can still produce sparse traffic. A dense *active connection* graph, frequent revocation, or sustained denied traffic can remove that advantage. This distinction supplies testable hypotheses without assuming a particular technology must win.

5 Implementation

The inspected implementation is the Go-based kube-network-policies repository at revision 9984dcd. Its dataplane controller uses nftables through netlink and the NFQUEUE interface for packet delivery and verdicts. The queue is configured with a maximum length of 1,024, a copy length of 128 bytes, GSO support, and a 100-ms write timeout. These are implementation settings, not measured service rates or tail latency bounds. Limiting the copy to headers avoids copying whole payloads for policy decisions, but requires checking parser behavior for options, extension headers, fragmentation, and truncation.

The callback resolves policy through an evaluator interface and emits an acceptance conntrack label with the verdict. Metrics record callback processing time, protocol/family-specific verdict counts, queue occupancy, and drop counters. Callback time does not include every component of kernel-to-userspace waiting time, so it cannot alone represent application-visible connection setup. The benchmark harness did not scrape these agent metrics: its Prometheus queries cover node and kubelet CPU only, and a `kindnetd` CPU query in the saved configuration returned no samples. The implementation also has host-traffic and protocol exceptions; the conceptual cached path in Figure 1 is not a complete description of all hooks.

The repository builds several flavors of the agent. The standard flavor queues only traffic to or from local Pods selected by some policy, maintained as nftables sets of managed addresses. The IPTracker flavor inspected at the same revision instead reports *divert-all*: it queues every forwarded packet not already bypassed by conntrack, because it watches only

NetworkPolicy objects and has no Pod informer from which to enumerate the local Pods a policy selects. The two flavors therefore have different λ_q for the same workload, and the queue-overload analysis of Section 3 applies to a larger traffic fraction in the IPTracker flavor.

The NRI resolver is an alternate source of local Pod metadata, not a replacement for every informer. Its callbacks add and remove local records; namespace information can still come from the API-derived view. This modularity permits policy logic to evolve as ordinary userspace code while the steering mechanism remains relatively stable. We regard reduced feature-development coupling as an architectural property, not a measured productivity result. New policy features may still require additional metadata, hook coverage, or cache invalidation.

An experimental distinction is essential: KNP names the inspected policy engine, while the benchmark directories label complete networking configurations as *Kindnet* or *Cilium*. Those configurations include networking and runtime components beyond policy evaluation. The benchmark repository pins Cilium v1.20.1 in its cluster specification and a kube-network-policies:v1.1.0 image in its standalone install manifest, but the KNP cluster was created with the Kindnet CNI’s bundled policy agent, patched to kindnet:v1.0.1 with an NRI socket mount; which policy binary and flags were effective in each Kindnet run is not recorded in the artifacts. We therefore do not attribute their latency differences solely to the code described here.

6 Evaluation

6.1 Methodology and Provenance

We analyze archived ClusterLoader2 artifacts rather than execute a new cluster experiment. The repository describes a testbed with 720 n2-standard-4 workers, a c4-standard-96 control-plane node, and a target of 35,000 concurrent Pods. The runtime configuration is reported as containerd 2.2.4 with runc 1.3.5. These are environment descriptions, not a reconstructed per-run inventory; topology and binary provenance must be confirmed before pooling runs from different dates.

The extraction at benchmark revision e8af592 inventories 1,809 artifact files and parses 1,285 JSON files, recovering 55 startup reports. Including diagnostic-only directories yields 62 attempts, seven without a startup report. Later revisions 0442dce and f998224 add descriptions of the raw-Pod abort and mesh-tier exclusion, respectively. Those records are used as documented outcomes, not as replacement timing samples. Appendix A identifies the relative source paths.

We distinguish create_to_schedule, schedule_to_run, and total pod_startup. The middle quantity includes runtime and container work, and in this harness it also includes a connectivity gate: every sandbox

Pod template carries a wait-for-gateway init container that loops `nc -z` against the gateway Service until a TCP connection succeeds. The sandbox is egress-isolated and the gateway is ingress-isolated by the deployed policies, so a Pod reaches `Running` only after one *permitted* connection has been admitted by both the source node’s egress evaluation and the gateway node’s ingress evaluation. `schedule_to_run` is therefore an upper bound on time to first permitted communication for the sandbox-to-gateway path, and it includes propagation of the new Pod’s address to the gateway node. It does not test sandbox-to-sandbox permissions, exercises no forbidden path, and under KNP’s fail-open default would also pass if the queue overflowed. The init container logs its own wait as `Latency: N ms`; those per-Pod values were not collected. Component percentiles cannot be added to reconstruct a total percentile. Likewise, a configured object-operation QPS is not necessarily achieved Pod throughput: a ReplicaSet operation can create multiple Pods.

Negative enforcement was checked once per cluster, not per run: `scripts/verify-netpol.sh` creates a client and server, confirms connectivity, applies a default-deny ingress policy and confirms the request is blocked, then applies an explicit allow and confirms it succeeds. Its output is not archived, and it ran on an idle cluster before the sweeps. The benchmark therefore establishes positive enforcement at scale and negative enforcement only as a quiescent smoke test.

Saved configurations take precedence over the current generator. The September identity sweeps vary Pods per ReplicaSet along with intended identity cardinality, use the default scheduler, and replace 5,000 Pods per churn round. Five rounds within a run are not five independent trials. The number of allocated Cilium identities must also be distinguished from the generator’s intended label cardinality. We retain per-run percentiles and completion indicators, rather than constructing CDFs or confidence intervals from summary quantiles.

6.2 Workload Admission and Identity Turnover

Table 1 reports completed September identity tiers across Cilium (open gateway, hub-and-spoke, and bidirectional mesh) and KNP (bidirectional mesh). All target 35,000 Pods at 500 QPS. Within the ReplicaSet-paced tiers from 700 identities upward, increasing intended identities does not produce an increasing post-scheduling tail for either engine. In Cilium’s open-gateway case, P99 decreases from 5.930 s to 1.990 s between the endpoint tiers (hub-and-spoke: 6.370 s to 2.033 s). In the bidirectional mesh, KNP P99 falls from 5.331 s at 700 identities to 2.791 s at 3,500 (P50: 3.273 s and 2.234 s), within 2.4 s and 0.5 s of Cilium’s 2.934 s and 2.279 s on the same generator.

The two remaining KNP tiers are not flat. At 7 identities

Intended identities	Pods/RS	Cilium P99 (s)			KNP Mesh P99
		GW	Spoke	Mesh	
7	5,000	5.930	6.370	6.767	151.479
700	50	5.416	4.759	2.934	5.331
3,500	10	4.685	4.899	2.279	2.791
35,000	1	1.990	2.033	Excl.	—
35,000	Raw	—	Abort [†]	—	421.410*

Table 1: Per-run archived schedule-to-run P99. All ReplicaSet-paced suites have zero JUnit failures. [†]Cilium raw-Pod hub-and-spoke run aborted at 29,306 Running / 5,694 stranded. *KNP raw-Pod mesh run: 35,000 Running, 0 stranded, two JUnit failures against the 5-minute P90 startup SLO (observed 357.7 s). The two raw-Pod runs use different policies and are not a matched pair.

with 5,000 Pods per ReplicaSet, KNP records 38.946 s P50 / 151.479 s P99 where Cilium records 1.41 s / 6.767 s under the identical configuration: a 22 $\times$ gap in P99 that the artifacts do not explain. It is consistent with the July Kindnet baseline (57 s P99) and tuned run (9.7 s post-scheduling, 95.8 s total) at the same 5,000-Pods-per-ReplicaSet shape, so the large-ReplicaSet burst, not identity count, is where the KNP configuration degrades. The 35,000-identity tier was run as 35,000 raw Pods under the mesh policy; it has no Cilium counterpart, since Cilium’s producer excluded the mesh tier analytically and Cilium’s only raw-Pod run used the hub-and-spoke policy and aborted. KNP completed it with all 35,000 Pods Running and no stranded Pods, at 120.003 s P50 / 421.410 s P99 post-scheduling and 139.6 s / 499.9 s end-to-end; the harness recorded two failures against its 5-minute P90 startup SLO. `create_to_schedule` reached 106.7 s P99, so scheduling itself took well over a minute for the tail. Candidate causes for the post-scheduling tail include serialized sandbox creation in kubelet (about 49 Pods arrive per node), convergence of 35,000 new Pod addresses to the gateway nodes before their init containers can be admitted, and API-server watch latency; the kubelet CNI-latency and agent queries returned no samples, so the artifacts do not separate them.

These results prevent a simple claim that identity cardinality alone causes Cilium startup collapse. They do not prove identity-independent scaling either: the generator changes controller fan-out and burst size with each tier. Table 2 makes the confound visible. Achieved Phase-1 scheduling throughput is 277–328 Pods/s (median of per-second samples) for every Cilium tier with 10 or more Pods per ReplicaSet, and 169–170 Pods/s for the two 35,000-ReplicaSet tiers; cluster-scoped Pod LIST P99 latency rises from 14–19 s to 28.8 s in the same tiers. The low post-scheduling P99 at 35,000 identities therefore coincides with a slower, controller-limited creation stream, which reduces the burst each node absorbs. One Pod per ReplicaSet is not equivalent to a few large ReplicaSets at

Scenario	Identities	Pods/s P50 / P90	Node CPU (cores)	
			mean	P99
Gateway	7	284 / 377	0.52	0.71
	700	317 / 364	0.51	0.69
	3,500	316 / 372	0.54	0.69
	35,000	170 / 182	0.56	0.81
Spoke	7	0 [†] / 327	0.51	0.69
	700	323 / 398	0.61	0.86
	3,500	321 / 373	0.54	0.67
	35,000	169 / 181	0.55	0.78
Cilium Mesh	7	277 / 376	0.54	0.72
	700	302 / 351	0.54	0.67
	3,500	328 / 385	0.63	0.77
KNP Mesh	7	0 [†] / 310	0.52	0.66
	700	124 / 134	0.48	0.61
	3,500	103 / 106	0.45	0.56
	35,000 (Raw)	0 [†] / 97 [‡]	0.66	0.86
Kindnet QPS 500	~7	233 / 262	0.40	0.46
Kindnet tuned	~7	190 / 225	0.63	0.79

Table 2: Recovered Phase-1 scheduling throughput and worker-node CPU for the runs in Tables 1 and 3. CPU is the harness’s instantaneous node query on 4-vCPU workers, aggregated across nodes over the run; the Kindnet baseline row uses an earlier rate-based query and is not directly comparable. [†]More than half of the per-second samples were zero. [‡]Peak 841 Pods/s. No memory series exists; the six KNP agent queries added for the mesh sweep returned no samples because the deployed Kindnet build does not expose the agent’s metrics endpoint.

the same object-operation rate. A controlled identity-turnover test should hold these quantities fixed and independently vary label sharing and label freshness.

Worker CPU does not track intended identity count in either family: Cilium’s mean stays within 0.51–0.63 cores and P99 within 0.67–0.86 across tiers and scenarios, and KNP Mesh stays within 0.45–0.66 mean cores (0.56–0.86 P99). This is a whole-node aggregate that includes kubelet, runtime, and the sandbox workload; it cannot separate the policy agent’s share, and a per-Pod policy-map dump was not collected. What it does exclude is a CPU explosion at 35,000 identities on the nodes: if Cilium pays for identity cardinality in these configurations, it does so in the operator, the API server, or in map capacity, not in visible worker CPU.

The mesh tiers are the only place the two engines meet on the same generator and policy. For Cilium, post-scheduling medians remain approximately 1.41–1.42 s across the three executed tiers, while end-to-end P99s are 33.534, 28.739, and 21.730 s (initial setup plus wait durations: 192.109, 137.947, and 109.718 s). For KNP at 700 and 3,500 identities, post-scheduling latency is 3.273 s P50 / 5.331 s P99 and 2.234

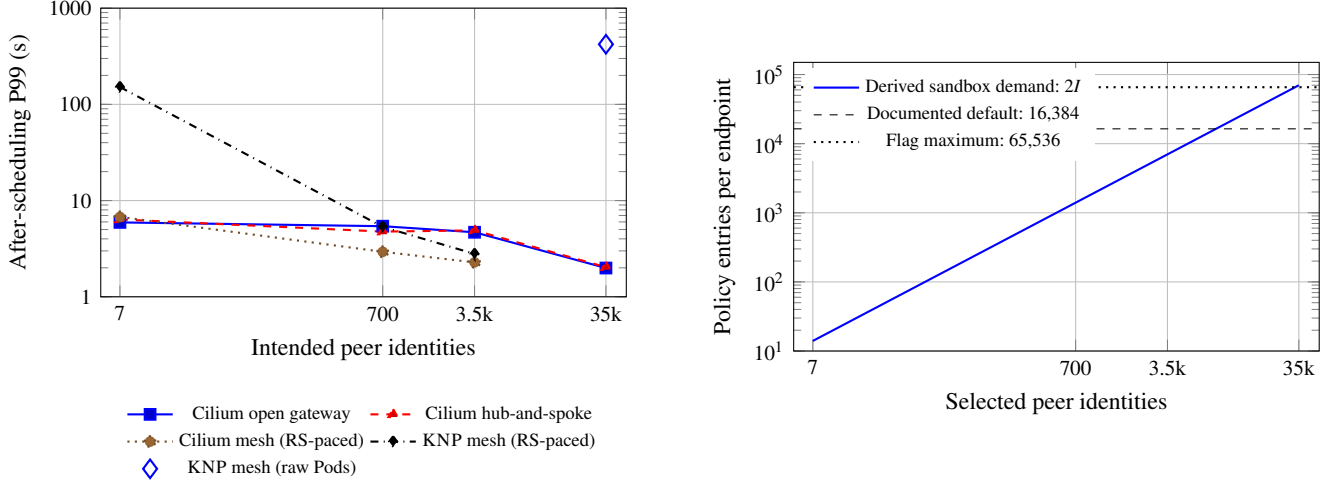


Figure 2: Different meanings of scaling. Left: measured per-run startup quantiles on a logarithmic axis so that every executed tier is visible; connecting lines are visual guides, and the raw-Pod point has no matched Cilium run. Right: conditional permission-state model for eager per-endpoint compilation. The runner documents `-bpf-policy-map-max` as 65,536 on the Cilium cluster, which admits the unidirectional 35k tiers and excludes the bidirectional mesh at 35k ($2I = 70,000 > 65,536$).

s P50 / 2.791 s P99, and end-to-end P99 is 5.553 s and 2.918 s, far below Cilium’s end-to-end values because KNP’s `create_to_schedule` P99 was 0.4 s and 0.2 s. Two confounds prevent reading this as an engine comparison. First, the KNP runs achieved 124 and 103 Pods/s median Phase-1 throughput against Cilium’s 302 and 328 on the same configured 500 QPS and identical control-plane settings; each KNP node therefore absorbed Pods at well under half the rate, which by the paper’s own argument lowers the per-node burst and the tail. The cause of the slower creation stream on the Kindnet cluster is not recorded. Second, each cell is a single trial. What the matched tiers do support is narrower: under a generator that Cilium handles at 2.3–2.9 s P99, KNP handles 700 and 3,500 identities at 2.8–5.3 s P99 with the gateway-admission gate included, and its worker CPU does not rise with identity count.

Version matters as much as cardinality. An archived Cilium v1.18.6 attempt at the 700-identity gateway tier, run three hours before the v1.20.1 sweep on the same day, retains only its generated configuration and metadata: the harness never produced a startup report. The repository’s narrative records 34,981 Pods Running and 19 stranded in `Init:0/1` under default deny, and attributes the outcome to a policy-cache race between identity allocation and endpoint regeneration tracked as Cilium issue 7515 and resolved in v1.20.0 [1]. We have not reproduced that diagnosis from logs. The point for this paper is narrower: the same generator, at the same tier, produced an admission failure on one release and a clean 5.4-s P99 on the next. Identity-scaling claims about eager-materialization systems are release-specific, and a comparison must pin and report the version it tested.

6.3 Capacity Exclusion Is Not a Latency Sample

The mesh experiment producer explicitly excluded the 35,000-identity tier on Cilium. Its model assigns an entry for each sandbox identity in both directions at every endpoint. Equation 1 then gives approximately 70,000 sandbox entries per endpoint, before gateway, DNS, or other entries. This exceeds the stated 65,536-entry capacity. The runner lists only the three smaller tiers for Cilium and contains a guard that skips estimated demands at or above the configured map limit.

This is meaningful design evidence: a system can have low latency throughout its feasible region and still encounter a state-capacity boundary. It is not evidence that an executed 35,000-identity mesh on Cilium experienced insertion failures, packet loss, or a particular latency. We label it an *analytical exclusion*. The effective `-bpf-policy-map-max` on the Cilium cluster is not archived: the runner reads it from the `cilium-config` `ConfigMap` at run time and falls back to the documented default of 16,384, and its header records the tested value as 65,536, the flag’s maximum [14]. The completed runs corroborate the higher value independently. The gateway policy admits ingress from every sandbox identity, so at 35,000 identities each gateway endpoint needs about 35,000 entries; behind a 16,384-entry map, Pods beyond that count could not pass their `wait-for-gateway` gate and would not have reached Running. All 35,000 did, in both unidirectional 35,000-identity tiers, so the effective limit was at least 35,000 unless v1.20.1’s identity aggregation applied, which the source restricts to cluster-, world-, and node-level entity buckets rather than label selectors. A `cilium-dbg bpf policy get dump` on one gateway node would settle the

value; none was taken. The agent also exposes an endpoint-lockdown-on-overflow option [14], so an executed overflow would surface as denied traffic rather than as a startup latency, which is one more reason the unrun tier has no P99.

Under the benchmark model, 3,500 selected identities and 50 local endpoints imply approximately 350,000 sandbox policy entries per node. This is derived demand, not a map dump or a memory measurement. The contrast with KNP is structural: its inspected design does not require this permission cross-product, so no per-endpoint capacity excluded its 35,000-identity mesh tier, and that tier completed with every Pod Running. Its capacity limits lie elsewhere—metadata storage, conntrack sizing, and userspace decision rate under connection establishment—and the 421 s tail of that same tier shows that removing the map bound does not by itself make the tier cheap.

6.4 An Aborted Direct-Burst Run

The raw-Pod microsegmentation record documents a different outcome. An experiment creating Pods directly at the configured 500-QPS tier aborted with 29,306 Running and 5,694 stranded out of a 35,000-Pod target. The record attributes the abort to local Cilium REST API rate limiting during concurrent CNI ADD calls and exhaustion of the 65,280 allocatable cluster-local identities across creation and churn. It reports the allocator message “no more available IDs in configured space.” No final latency results were generated.

The missing percentile file is not a reason to erase a failed run. However, the documented diagnosis is not the same as a reconstructed event timeline. Allocator history, garbage collection, effective limiter settings, and the underlying logs are needed to separate the two causes and determine their interaction. Historical allocations can exceed the live-Pod count, but turnover alone does not imply monotonic consumption of an unreclaimable identity space.

The allocator namespace here is cluster-level state. It is distinct from the per-endpoint policy-map capacity used to exclude mesh tier 4. Likewise, direct Pod bursts differ from ReplicaSet-mediated creation. The raw-Pod abort supports an admission-failure case study, not an assertion that every 35,000-identity workload fails or that the unrun mesh tier has a measurable failure P99.

6.5 Connection Cost and Correctness

The available collection measures workload startup and associated control-plane activity, not the core packet-admission tradeoff. It does not establish NFQUEUE verdict capacity, established-flow throughput parity, or isolation through startup and failure. It does bound first-permitted-connection latency, because the `wait-for-gateway` gate of Section 6 folds one admitted connection into every `schedule_to_run` sample, but it does not isolate that component from image

pull and container start. A configured mesh permission does not create all-to-all traffic, and reaching Running does not test any forbidden peer relationship; the only negative check is the one-shot smoke test described there.

The only connection-level data point is the repository microbenchmark summarized in Section 4: 2,317 new connections/s through the queue on one node with sub-millisecond callback medians, and a P99 above 3 s once conntrack was exhausted [18]. Read against the startup sweeps, it says that a node absorbing 50 sandboxes at 170–330 Pods/s cluster-wide is nowhere near the queue’s connection limit, and that the limit is reached first by connection-tracking capacity rather than by the userspace evaluator. It does not say what happens under sustained denied traffic, whether the P99 is queueing or conntrack, or how the numbers change with policy count.

The minimum discriminating extension is a small, controlled comparison of fresh and reused TCP/UDP flows at increasing offered rates, with permitted and denied traffic kept separate. It should report achieved connection rate, application-visible P50/P99 setup latency, queued packets per successful connection, agent CPU, queue drops, and conntrack occupancy. Repeated trials should include overload, rather than stop before the queue becomes the bottleneck, and should run once with fail-open and once fail-closed to measure the bypass window of Section 3. This can test the displaced cost without repeating a full 720-node sweep.

A companion lifecycle test should establish policy before Pod creation, then observe runtime metadata delivery and both allowed and forbidden connectivity from init containers onward. An NRI ablation needs otherwise identical versions and fairness settings. Policy changes, agent restart, queue overflow, and IP reuse are required to delimit the safety claim. These are remaining evaluation requirements, not experiments whose results this draft assumes.

6.6 Control-Plane Sensitivity and Limitations

The July QPS experiments illustrate why startup must be decomposed. Table 3 reports primary saved Cilium and Kindnet baseline runs. At QPS 500, their post-scheduling P99s are 5.482 s and 57.068 s, respectively. The archived baseline therefore does not establish a general startup advantage for the NFQUEUE-based configuration. At QPS 50 and 100, both families record two JUnit failures. All four are the same measurement, `WaitForOldPodsDeleted-Final`, timing out while 5,024 to 19,999 sandbox Pods from the churn phase were still terminating; no startup-phase measurement failed. The startup percentiles in those rows are therefore complete, but the runs did not finish their teardown within the harness deadline, and Pod deletion latency at these tiers is itself unmeasured.

The final Kindnet tuning bundle, labeled version 1.0.1 with NRI and APF exemption, records a post-scheduling P50 of 2.345 s and P99 of 9.718 s. Its total P99 is nevertheless 95.790

QPS tier	After scheduling P99		Total P99	
	Cilium	Kindnet	Cilium	Kindnet
50*	4.065	4.833	4.089	4.855
100*	1.788	7.143	1.814	7.378
200	1.845	59.347	2.285	59.485
500	5.482	57.068	11.160	57.629

Table 3: Archived primary QPS runs; all values are seconds. Asterisks mark tiers whose only JUnit failures are final-teardown deletion timeouts in both configurations; startup measurements completed. These are complete-stack observations, not an isolated NFQUEUE versus eBPF comparison.

s, with a create-to-schedule P99 of 94.555 s. This juxtaposition is evidence against interpreting a reduced post-scheduling tail as an automatic end-to-end speedup. Versions, runtime integration, and API fairness settings changed together, so the sequence is not a causal NRI ablation. A separate Cilium QPS-500 repetition records 4.257 s post-scheduling P99 and 12.404 s total P99; it is retained as a separate trial.

API Priority and Fairness (APF) divides API-server concurrency among request classes and provides queueing and isolation [16]. Increasing one class’s share or exempting it can shift contention to other components; it does not create storage, CPU, or network capacity. The API server also mediates most declarative control-plane changes, including authorization, admission, persistence, and watch delivery. Its central role makes API latency relevant to network-policy evaluation, without implying that application packets pass through it.

The collection’s limitations affect the conclusions directly. Most configurations have too few independent trials for statistical claims. Summary percentiles hide time variation and exclude reconstruction of individual Pod trajectories. The counts of failed JUnit elements are not counts of failed Pods. CPU maxima based on instantaneous samples at the harness’s scrape interval are not true peaks. We use these artifacts to locate the questions the architecture must answer, not to manufacture a comparative scaling curve from missing measurements.

6.7 What the Artifacts Still Do Not Contain

The KNP mesh sweep closes the largest gap of the earlier draft: a KNP run above seven identities now exists at 700, 3,500, and 35,000. It does not close the comparison. The harness added six KNP agent queries (verdict rate, processing latency, queue and user drops, CPU, memory) and a kubelet CNI-latency query, and every one returned no samples in all four tiers; the repository’s own probe script concludes that the deployed Kindnet build has the agent metrics endpoint disabled. Table 4 lists what remains. Each row names the missing evidence, the claim it would support or refute, and the cheapest experiment that would produce it.

Missing evidence	Claim at stake	Cheapest producer
Agent CPU, memory, verdict rate, latency, drops (all six queries empty)	where displaced work lands; λ_q, μ , fail-open window	rebuild Kindnet with the metrics server enabled; rerun one tier
Cause of the KNP 151 s and 421 s tails	burst behavior of the KNP node path	collect init-container Latency: logs and kubelet PLEG/CNI metrics on a sample of nodes
Why KNP runs created Pods at 103–124 Pods/s versus Cilium’s 302–328	validity of the matched-tier comparison	compare controller-manager and API-server metrics of the two clusters
A Cilium raw-Pod mesh run or a KNP 1-Pod-per-ReplicaSet 35k run	a matched pair at 35,000 identities	one additional run on either cluster
Repeat trials	any statistical statement	rerun tiers 2–3 on both clusters
Policy-map occupancy dump and the effective map-size flag	capacity model of Fig. 2	cilium-dbg bpf policy get and the ConfigMap on one node
Denied traffic under load	isolation, deny amplification	single-node connection microbenchmark with deny mix, fail-open and fail-closed
NRI on/off with identical binaries	startup-metadata benefit	rerun baseline QPS 500 and tuning bundle 1 with one pinned image
Binary digests and flags per run; logs for both aborts	attribution and causal diagnosis	record in c12-metadata; retain logs on abort

Table 4: Evidence still absent from the benchmark and implementation repositories after the KNP mesh sweep.

7 Related Work

The closest precedent is reactive virtual switching, not another use of a kernel packet API. OVS combines userspace classification with kernel flow caches and discusses the performance consequences of that architecture [9]. NVP addresses distributed logical network realization [4]; OVN supports stateful ACLs [23]. Our proposed distinction is evaluation of Kubernetes policy semantics without eager per-endpoint peer materialization, coupled to local runtime metadata. We do not claim the invention of userspace miss handling, cached verdicts, or stateful filtering.

Andromeda separates flow-processing paths to reconcile provisioning, isolation, performance, and feature velocity [2]. ONCache identifies repeated work across container-overlay layers and caches invariant results, including filtering, using eBPF [5]. These systems reinforce the importance of choosing what to cache and when to invalidate it. KNP focuses on

the timing of semantic policy work, whereas ONCache accelerates repeated overlay processing. Its microbenchmark and application evaluations also illustrate the evidence missing from a startup-only comparison.

KUBEDIRECT identifies API-mediated controller communication as a serverless scaling bottleneck and develops state management for a more direct path [10]. It is relevant to our distinction between local runtime observation and eventual publication of state, but NRI-based address discovery does not provide KUBEDIRECT’s broader controller coordination. Role-based micro-segmentation infers endpoint roles and policies from traffic [6]; this addresses policy construction rather than enforcement of a given policy.

Finally, metadata aggregation is established practice. Calico Typha maintains datastore state, deduplicates events, and distributes updates to node agents to reduce their individual load on the datastore [25]. The IPTracker direction discussed next must therefore be evaluated on its representation, caching, and recovery tradeoffs, not on the novelty of inserting a distribution tier.

The eager-materialization model of Section 4 is one representation, not the only kernel-resident one. Open vSwitch’s conjunctive match (`conj_id`) lets a rule with n source and m destination members be installed as $n + m$ flows rather than $n \times m$ [22], and Antrea’s OVS pipeline uses exactly this to compile Kubernetes NetworkPolicy, marking established connections in `conntrack` so that only new flows traverse the policy tables [13]. Calico’s Felix compiles selectors into ipsets referenced by a bounded number of iptables or nftables rules, so per-endpoint state grows with the number of *rules* rather than peers [24]. Both approaches keep semantic evaluation in the kernel and avoid the per-peer cross-product by factoring the match; KNP avoids it by moving the match to userspace and caching the result. A fair capacity comparison must therefore include a factored-kernel baseline, not only the per-identity expansion that motivated the mesh-tier exclusion.

8 Discussion and Future Work

8.1 Compact Metadata Distribution with IPTracker

Avoiding a distributed security-identity allocator does not eliminate distributed metadata. With N agents each watching all Pod updates, an update stream of rate U entails approximately NU deliveries under an all-subscriber model. This is a first-order count, not a complete performance model; filtering, watch caching, event contents, and implementation details determine the actual cost.

IPTracker, introduced in the implementation’s PR 190, moves full Pod observation to an intermediate set of distributors and exposes compact IP-to-Pod records to agents [7]. If R distributors serve N nodes, the policy agents’ full Pod subscriptions against the API server can fall from N to R .

Other subscriptions remain, and the distributor-to-agent fan-out does not disappear. In the inspected client, each agent watches the entire compact-record prefix. This is not selective subscription to only the peers an agent contacts.

The records are serialized with Protocol Buffers and may be stored locally in `bbolt`, with an in-memory LRU in front [17]. Kubernetes can itself use `protobuf`, so the relevant distinction is the reduced schema and distribution architecture, not a blanket JSON-versus-protobuf comparison. The disk store also records revision metadata for synchronization. A cache miss can consult local storage rather than perform a per-packet network lookup.

Locality offers a potential memory benefit: a node may communicate with a small fraction of the cluster’s endpoints even when it stores metadata for all of them. However, a bounded object cache is not a bound on process or machine memory. `bbolt` mappings and page cache, full-sync response buffers, and whole-store listing contribute memory cost. The present LRU admits updates as well as lookup results, so metadata churn can evict frequently contacted peers. Store access under the wrapper’s mutex can also couple writes and lookup latency.

Recovery and freshness are equally important. The current full sync fetches the prefix, clears local state, and inserts records; this does not establish atomic snapshot replacement or constant-memory bootstrap. Revision recovery, compaction, reflector failover, disk failure, and IP reuse must be evaluated before persisted state is treated as current authorization metadata. Issue 114659 provides additional motivation for reducing informer memory costs, but it is not a quantitative demonstration of this implementation’s benefit [3].

The only quantitative data on IPTracker are the single-client integration-test logs posted with the repository’s scalability test (PR 218) [8]. With 10,000 records the server reported 35 MB of Go heap, 643 ms average propagation delay, and a 1 MB client store. A 40,000-record run at 30,000 writes/s reached 6.6 s propagation and 174 MB, and the `etcd` maintainer participating in the thread observed watch propagation stalling above that size. A 10-million record run throttled to 1,024 writes/s completed in 9,828 s with 43 ms average propagation, 24.7 GiB of server heap, and a 1.25 GiB client store. These are one process on one machine with no packet path, and the LRU in that test was sized to hold every record, so they bound neither agent memory under eviction nor the fan-out to hundreds of clients. They do show that the distribution tier, not the client, holds the bulk of the state, and that write rate rather than record count governs propagation delay in the tested range. IPTracker is implemented future work in this paper’s evaluation scope, not a feature credited with the reported startup results.

8.2 Testing the Architectural Hypothesis

The most informative next result is not necessarily a larger cluster. First, effective software versions, identity counts, map limits, and the source of each plotted result must be fixed. Existing lower-tier map dumps could validate the capacity model without executing a tier already excluded by the harness. Existing failure logs could establish the allocator and limiter timeline without reproducing an expensive abort.

Second, workload cardinality and workload freshness should be varied independently. A matched experiment can retain the same live-Pod count, generator, placement, offered rate, and policies while alternating between replacements that reuse label sets and replacements that introduce new ones. Measuring first permitted communication alongside startup and actual map occupancy would distinguish distribution and materialization from unrelated controller delay.

Third, the connection-cost experiment must determine when deferred evaluation stops being advantageous. If accepted-flow reuse is low, denied demand is high, or strict revocation repeatedly invalidates cached decisions, the userspace path may dominate. These outcomes would delimit the design’s useful region, not invalidate the use of NFQUEUE for every workload. Likewise, a demonstrated capacity benefit would not establish better packet throughput or immunity to control-plane overload.

8.3 Limits of the Present Claim

The design does not promise unlimited identities, constant total memory, zero startup delay, or topology-independent resource use. It avoids one form of eager permission expansion by retaining semantic interpretation in userspace. This is a useful separation only if the resulting admission and consistency costs are acceptable. The KNP mesh sweep shows that a tier excluded for Cilium by per-endpoint capacity completes under KNP, and that 700 and 3,500 identities cost KNP no more than seven do; it also shows a 151 s tail under a 5,000-Pod-per-ReplicaSet burst that Cilium absorbs in 6.8 s, at lower achieved throughput and with one trial per tier. The benefit is therefore demonstrated for capacity and not yet for admission latency, and the packet-level boundary remains for the measurements of Table 4.

9 Conclusion

Network-policy scalability is a joint problem of state representation, work placement, and synchronization. Fast kernel lookups do not guarantee cheap workload admission, and a flat latency curve does not imply unbounded policy capacity. NFQUEUE and conntrack provide an existing foundation for a different decomposition: interpret Kubernetes policy when communication is attempted and retain accepted connection decisions in the kernel. NRI can remove the asynchronous

API round trip from local Pod-address discovery without pretending to eliminate all distributed-state dependencies.

The archived experiments on 35,000 Pods and 720 workers show both sides of this trade-off. Completed Cilium sweeps retain low post-scheduling latency and flat worker CPU within their feasible region, halve achieved throughput under 35,000 ReplicaSets, exclude the 35,000-identity bidirectional mesh on predicted map capacity, and stranded 5,694 Pods in one hub-and-spoke raw-Pod burst. The KNP mesh sweep completes that excluded tier with every Pod Running, matches Cilium within 2.4 s P99 at 700 and 3,500 identities, and records tails of 151 s and 421 s in its two burst tiers that the artifacts cannot yet attribute. Because every sandbox gates on a permitted connection to the gateway, these latencies include first permitted communication across nodes, which makes them more informative than plain container startup and also makes the unexplained tails more important. Completing the comparison requires the measurements of Table 4, starting with an agent build that exposes its metrics and a repeat of the matched tiers.

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A Evidence Provenance

This appendix is an internal provenance guide for the non-submission exercise. Paths are relative to the benchmark repository unless stated otherwise. It is not anonymized publication material.

Archived numeric results. The extractor is `submission/analyze.mjs`; the current extraction (covering all 66 benchmark runs and 59 startup reports) is recorded in `submission/upstream/inventory.json`, `submission/upstream/startup.csv`, and `submission/upstream/extracted-results.md`. Run names do not independently establish tested software versions.

Table 1 and the measured panel of Figure 2 use reports beneath `artifacts_pods/cilium/identity-sweep/`, `artifacts_pods/cilium/microsegmentation/`, `artifacts_pods/cilium/mesh-sweep/`, and `artifacts_pods/kindnet/mesh-sweep/`. The Cilium mesh reports are in `tier1-7-identities-5000ppr`, `tier2-700-identities-50ppr`, and `tier3-3500-identities-10ppr`. Their sibling `*-initial` directories contain hook logs, not startup reports. The KNP (Kindnet) bidirectional mesh reports are in `artifacts_pods/kindnet/mesh-sweep/tier1-7-identities-5000ppr`, `tier2-700-identities-50ppr`, `tier3-3500-identities-10ppr`, and `tier4-35000-identities-rawpods`. Their `generatedConfig_agentic-sandboxes.yaml` files set `SandboxPeerMode: mesh`; the Cilium raw-Pod directory under `microsegmentation/` does not, which is why the two raw-Pod runs are not a matched pair. The `tier-4 junit.xml` records the two startup-SLO failures. The `GenericPrometheusQueryKubeNetworkPolicies*` and `...KubeletCNIPluginOperationsLatency` files in all four tiers contain `"dataItems": null`; the queries are defined in `manifests/agentic-sandbox/measurements/knp-measurements.yaml` and `scripts/query-kindnet-metrics.sh` records that the Kindnet build exposes no metrics port.

Enforcement gate. The `wait-for-gateway` init container is defined in `manifests/agentic-sandbox/manifests/sandbox-pod.yaml` and `sandbox-replicaset.yaml`; the policies that make the gate a permitted-connection test are `global-sandbox-policy.yaml` (`sandbox egress to group: gateway`) and `gateway-policy.yaml` (`gateway ingress from group: sandbox`), with `default-deny-policy.yaml` applied in every archived run (`ApplyDefaultDenyPolicy: true`). The one-shot negative check is `scripts/verify-netpol.sh`; no output of it is archived.

Table 3 uses the primary runs under `artifacts_pods/cilium/qps-*`, including `qps-500/verified-run`, and `artifacts_pods/kindnet/baseline-sweep/qps-*`. The separate Cilium trial is in `qps-500/repeat-run`. The final Kindnet tuning bundle is under `artifacts_pods/kindnet/500qps-optimizations/5-v1.0.1-plus-nri-plus-apf-exempt/`. The `teardown-timeout` diagnosis of the asterisked rows is from the `junit.xml` files in the QPS 50 and 100 directories of both families.

Recovered throughput, CPU, and API latency. Table 2 reads `Phase1SchedulingThroughput*.json` (`perc50`, `perc90`) and `GenericPrometheusQueryWorkerNodeCPUUtilization*.json` (`mean_`, `p99_instantaneous_node_cpu_cores`) in each run directory; the Kindnet baseline uses `mean_node` and `p99_node` from an earlier query. Pod LIST latency is the cluster-scoped `Pods/LIST` row of `APIResponsivenessPrometheus_simple*.json`. The hub-and-spoke rows come from `artifacts_pods/cilium/microsegmentation/tier*`. The extraction is reproducible with `submission/analyze.mjs`; no value was transcribed from the repository’s Markdown summary.

Archived v1.18.6 attempt. `artifacts_pods/archive/v1.18.6-identity-runs/` holds a complete 7-identity run and a 700-identity directory containing only `cl2-metadata.json` and the generated configuration. The 34,981/19 counts and the issue-7515 attribution are from `benchmark_results.md`, not from artifacts.

Outcome records and capacity model. Revision 0442dce adds `artifacts_pods/cilium/microsegmentation/tier4-35000-identities-rawpods/README.md`, the source of the 29,306/5,694 completion counts and stated diagnosis. Revision f998224 adds `artifacts_pods/cilium/mesh-sweep/README.md`, documenting the unexecuted Cilium mesh tier. Revision 5b152f2 adds `artifacts_pods/kindnet/mesh-sweep/three ReplicaSet-paced KNP tiers matched to the Cilium mesh generator and one raw-Pod tier`. Its README’s “under 50 seconds” scheduling statement is contradicted by the archived `create_to_schedule P99` of 106.7 s, and its “0 CNI errors” has no backing metric; neither is used here. The model and exclusion are cross-checked against `scripts/run-mesh-sweep.sh` and `manifests/policy-scenarios/scenario-c-mesh-bidirectional.yaml`. The 65,536 map limit is documented only in the runner’s header comment; the ConfigMap it reads at run time is not archived.

Original charts. The input figure `figures/fig2_identity_cardinality_sweep.png` mixes measured quan-

tiles, unmatched Kindnet tier values, and a failure bar. The separate figures/fig2_topological_scaling_p99.png contains a 94.3-second overflow value and a kernel-set description of KNP that the audited evidence does not support. Both originals are preserved; neither is reproduced as a verified result here. Figure 2 is typeset directly from the audited quantiles and the explicitly conditional capacity model.

Implementation. Paths in the companion kube-network-policies repository are pkg/dataplane/controller.go (queue configuration, bypass flag, postrouting and input chains), pkg/cmd/cmd.go (-fail-open default), pkg/networkpolicy/networkpolicy.go (nil-peer selector semantics, managed-IP set), plugins/iptracker/iptracker_networkpolicy.go (divert-all), pkg/podinfo/nri/resolver.go, pkg/ipcache/{client,bbolt,lru}.go, and docs/testing/README.md with its two PNG charts, inspected at 9984dcd. Benchmark-side deployment facts are from manifests/kube-network-policies-install.yaml and scripts/create-netpol.sh. Source behavior has not been equated to the benchmark binaries. submission/evidence-audit.md records the claim-level audit, and submission/author-guide.md provides the prioritized remaining checks.